

Problem

Design a parallel resonant RLC circuit with $\omega_0 = 100$ krad/s and a bandwidth of 10 krad/s. Additionally what is the value of Q ?

Step-by-step solution

Step 1 of 3

Write the formula for bandwidth of a parallel RLC circuit.

$$B = \frac{\omega_0}{Q}$$

Substitute 100 krad/s for ω_0 and 10 krad/s for B .

$$\begin{aligned} Q &= \frac{\omega_0}{B} \\ &= \frac{100 \times 10^3}{10 \times 10^3} \\ &= 10 \end{aligned}$$

Hence, the quality factor, Q , of the circuit is 10.

Step 2 of 3

Write the formula for quality factor.

$$Q = \omega_0 RC$$

Substitute 10 for Q and 100 krad/s for ω_0 .

$$10 = 100 \times 10^3 RC \quad RC = 10^{-4} \dots\dots (1)$$

Write the formula for resonant frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 100 krad/s for ω_0 .

$$\begin{aligned} 100 \times 10^3 &= \frac{1}{\sqrt{LC}} \\ LC &= \left(\frac{1}{100 \times 10^3} \right)^2 \\ &= 10^{-10} \dots\dots (2) \end{aligned}$$

Step 3 of 3

In a parallel RLC circuit, the impedance is maximum and admittance is minimum.

To get minimum admittance value choose $R = 10 \text{ k}\Omega$.

Substitute $10 \text{ k}\Omega$ for R in equation (1).

$$RC = 0.1 \times 10^{-3}$$

$$(10 \times 10^3)(C) = 10^{-4}$$

$$C = 10 \text{ nF}$$

Substitute 10 nF for C in equation (2).

$$LC = 10^{-10}$$

$$L(10 \times 10^{-9}) = 10^{-10}$$

$$L = 10 \text{ mH}$$

Hence, the parameters of RLC parallel network are $R = 10 \text{ k}\Omega$, $L = 10 \text{ mH}$, and

$$C = 10 \text{ nF}.$$